RFmega

RF Generator

Operations Manual
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WARNING

TO REDUCE THE RISK OF FIRE OR ELECTRICAL SHOCK DO NOT EXPOSE THIS EQUIPMENT TO RAIN OR MOSITURE.

WARNING

Only qualified electrical workers should operate and install this system.
Avoid spilling liquids onto/into the unit.
Never operate without earth ground.
Do not expose to excessive heat or moisture.
Do not open – there are no user serviceable parts inside.
Do not block the chassis vent slots or the fan inlet.
Do not operate in an explosive environment.

The RFmega is a standalone RF driver designed to generate RF phase and anti-phase outputs. The RFmega is designed to drive high capacitive loads and also generate high RF voltage for lower capacitance loads. The capacitance of the load is resonated with the RF coil in the RFmega. The load capacitance and this coil inductance form a resonate circuit that defines the operating frequency. With high capacitive loads the RF currents can be very high and as a result it can require significant power to generate the desired voltages.

The RFmega comes in two different communication options:

- USB virtual RS232 interface
- Both Ethernet (192.168.1.50 default IP address) and USB interface

The features of the RFmega are listed below:

- Powered with 24VDC wall supply (120 watt), CUI SDI120-24-U-P51
- USB interface with ASCII control commands
- Compatible with MIPS host application
- Load capacitance range, 30pF to 4000pF
- Frequency range of 1MHz to 5MHz
- Phase and anti-phase outputs, 180 degrees phase shift between outputs
- User replaceable coils
- Output voltage range of 5 to 5000Vp-p, this depends on the load
- PL259 output connectors or SHV depending on application
- BNC output monitors
- DC bias inputs, +/-500V range
- Auto tune capability
- RF on indicator
- The system always powers up with the RF disabled
- Control TTL input, not implemented on this version
- No power switch, the system powers up when power is applied

This system has the ability to generate high RF voltages and high RF currents. Please be very careful in connecting to your system. If your load is unable to handle the high RF voltage and current you could damage your system and in extreme cases cause a fire.

- Never run the system unattended
- Never apply or remove the load when the system is running
- Never block the cooling holes
- Make sure you use heavy gauge coax to connect to your load, we recommend RG-213 for high capacitance loads.

The RFmega system is capable of communicating to the host using the USB interface or an optional Ethernet module. This Ethernet interface is configured using the host commands defined later in this document. This is a simple interface and does not support name registration or lookup so you need to connect to the IP address assigned to the interface.

The RFmega system will automatically look for the Ethernet module on power up and if found it will be initialized and enabled.

The MIPS host application supports connections on port 2015 so if you change the port number the MIPS app will not be able to connect. The MIPS app uses TCP/IP sockets to communicate. You can also make a telnet connection to the RFmega Ethernet interface and send any valid RFmega host commands.

Front Panel Controls:

24VDC input

A 24VDC 120 watt wall supply has been provided with the system. Never use a supply over 180 watts. Please note the 4 pin DIN connectors do not have universal wiring. Please only use the supply provided with the RF driver.

Power

This is a green LED that will light when the 24 volt power supply is connected. Note, the system will always power up with the RF output disabled. Even with the outputs disabled the system will generate a few volts of RF.

RS232 (for the model with RS232 interface)

The RS232 interface is designed to communicate with the MIPS host application. This interface has a fixed baud rate of 115,200. The commands are documented later in this document.

Ethernet and USB

Both a RJ45 ethernet and micro-USB interface are provided for the model that provides ethernet interface. Both interfaces can be used. The USB interface appears as a virtual com port on your PC and you should use 9600 or 115,200 baud.

Mode

The Mode button has two functions. If you press and release the button quickly the RF output will enable and disable. If you press and hold the button for 3 seconds the system will enter the Auto Tune mode, when the button is release after the 3 second hold. When the system is in Auto Tune mode the status LED will flash. This procedure takes about 1 minute to complete. The RF power level is set at 10% during Auto Tune.

Status

The Status red LED will flash when the system is in Auto Tune mode. This LED will signal additional information in future versions.

RF ON

This RF ON red LED is lit with the RF outputs are enabled.

Control

This opto isolated control input is not implemented on this version.

Bias

The bias input accepts a DC voltage from -500 volts to +500 volts and applies this DC to both RF outputs. This DC voltage is added to the RF voltage generated by the RFmega driver.

Rear Panel Connections:

Shown above are two rear panel options, the first option shows PL-259 connectors and is designed for high capacitance loads while the second option used SHV output connectors and is designed for high voltage applications. Your invoice we defive the specific configuration of your system.

The rear panel has the RF+ and RF- outputs as well as the monitor outputs. The RF+ and RF- are on PL-259 connectors or SHV. Your system was shipped with mating connectors that are designed for RG-213 cable when equipped with the PL-259 connectors. This is heavy duty cable and well suited for high power and RF current applications.

Two different types of monitors are used depending on your system.

High capacitance load system (PL-259 output connectors):

The monitors are capacitively coupled to the RF outputs using two 2200pF Mica capacitors in parallel. You can use a scope to monitor the RF on these outputs. Use a 10:1 or 100:1 scope probe. You can install BNC shorting capacitors on these BNC connectors to create a dummy load for the RFmega, this will allow you to test the system and provide a capacitive load similar to a SLIM board.

Note! The system is shipped with the sorting caps installed, you can test in this configuration but make sure and remove the shorting caps when testing with your actual load.

High voltage system (SHV output connectors):

In this configuration the monitor outputs are capacitively divided from the actual output voltage and are available on BNC connectors. These are intended for monitoring and accurate to within 5%. Use a short BNC cable or a 10:1 scope probe to monitor these outputs. The calibration of the monitor outputs are shown on the rear panel.

Grounding:

You will find a grounding lug on the rear panel. This lug is used to bond the RFmega chassis to earth ground. It is critical that this is the first connection made and the last connection removed when working with the RFmega.

Balance Adjustments:

Note: Balance adjustment is only possible on the high capacitive load version.

The balance adjustment allows you to adjust the capacitance to ground on the RF+ output to affect the balance. A perfectly balanced system will have the same RF level on the RF+ and RF- outputs. If the RF voltage RF+ and RF- do not match with in your requirements you can use the balance jumpers to make corrections. The bottom of the RFmega has a ¾" hole plug that can be removed to expose the balance jumpers. These jumpers enable capacitance to be added to the RF+ side of the RFmega to ground. The figure below shows the jumpers and lists the capacitance options depending on the number of jumpers installed. Increasing the capacitance will lower the voltage on the RF+ phase and raise it on the RF- phase. Changing the capacitance will also slightly change the resonate frequency. Adjust the jumpers to achieve the desired balance. Make sure the power is off before the jumpers are changed. Also make sure you save the RFmega parameters before you turn off the power.

Install jumpers to add capacitance from the RF+ output to ground. The capacitance for each jumper position is listed below. Multiple jumpers can be installed to increase capacitance.
1 = 10pF
2 = 20pF
3 = 47pF
4 = 82pF

Coil Replacement:

Note: Field coil replacement is only possible on the high capacitive load version.

The RFmega generator has user replaceable coil sets. The RFmega operating frequency is defined by the resonance frequency of its coils and the capacitance of your load. If

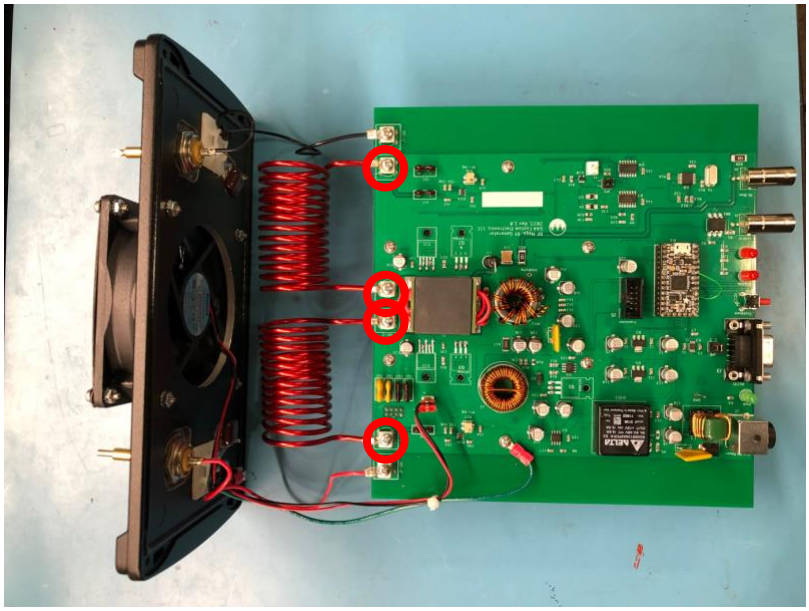
you need a different operating frequency the coils will need to be changed. This procedure outlines the steps to replace the coils.

1.) Remove all connections to the RFmega
2.) Remove the 4 torques screws that secure the rear panel, shown circled in red below:

3.) On the bottom of the case, you will find one Philips head 4-40 machine screw, remove this screw.
4.) Turn the RFmega on its side with the bottom facing you and the rear panel to your left.
5.) Carefully slide the internal PCB out of the case as you guide the rear panel. Please note the rear panel is connected to the PCB with a number of cables so it will not come free of the system. Below is a picture of the PCB sliding out of the case.

At the bottom of the PCB you can see the standoff that is used to hold the PCB in place. Use care when sliding the PCB out to make sure you do not damage this standoff.

6.) After the case is removed you can lay the PCB flat on your work bench with the heatsink setting down on the bench surface.
7.) The Coils are replaced in pairs. The picture below shows the coils and the 4 screws that secure the coils in place.



8.) Reverse this procedure to reassemble the RFmega.

Winding new coils

Supplies are included with the RFmega to allow you to wind your own coils. Please refer to the coil ship with your RFmega for details. You can find tools on the web that will allow you to calculate the needed inductance and then calculate the number of coil turns you will need.

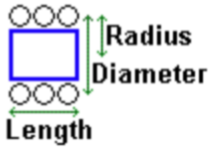
The resonate frequency is determined by the following equation:

$f = 1 / (2 * \pi * \sqrt{(L * C)})$
Where:
L = inductance is henries
C = capacitance in farads

The coil inductance can be calculated using the following equation:

(The following inductance formula requires units in inches.)

$$L = \frac{\text{Radius}^2 \times \text{Turns}^2}{9\,\text{Radius} + 10\,\text{Length}}$$



10 Gauge magnet wire and solder lugs are provided to wind the coil, shown below.

Calibration procedure

The RF mega uses a pricewise linear lookup table to convert the data from its level detectors into the displayed values. The system is calibrated in the factory but you many want to calibrate it in your specific application. The two output channels are calibrated separately. The system will need to be connected you your load and tuned to your normal operating conditions. You will need an oscilloscope to monitor the output voltages. You can calibrate using the voltage monitors on the rear panel or at any point in your system that you wish to control.

The RF mega will need to be connected to the MIPS host app and you will use the Terminal interface to issue commands and enter data. You will need to operate your system over the range you wish to calibrate. In this calibration procedure you will set RF drive levels and then enter voltage readings from your oscilloscope. Before you start the calibration first determine the set of RF drive levels you wish to use. Select a maximum of 8 points and a minimum of 4.

The calibration procedure will be run twice, once for each output phase.

The following steps outline the procedure:

1.) Enter the command *CALRFL,RF+* you will see the following message:

This function will generate a piecewise linear
calibration table. Set the drive level to reach
desired calibration points and then enter the
measured value to the nearest volt. Press enter
When finished. Voltage must be increasing!

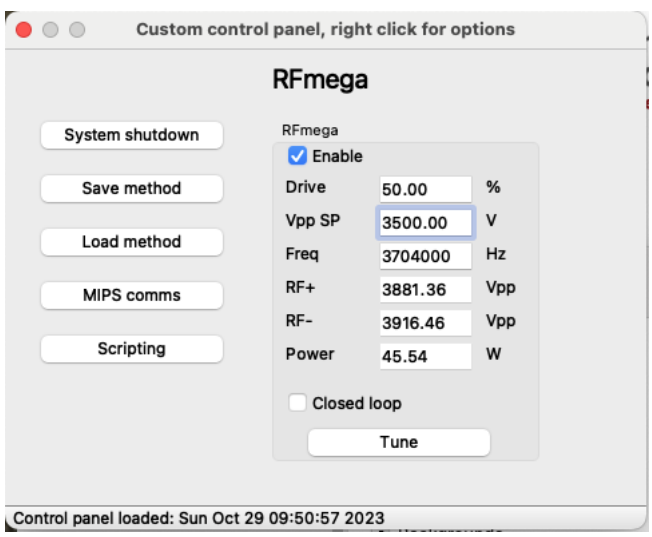
Point 1
Set drive level, %:

2.) Enter the RF drive level for this calibration point. Please note, this is a very simple interface and backspace is not supported. To exit this procedure press enter with no other characters and the system will exit and calculate the calibration table with the data entered.

3.) If you entered a drive level the system will set the drive to the value entered and then ask for the measured value, you will see the following message:
    Enter measured voltage:
4.) Enter the RF level measured with the oscilloscope at your calibration point. Press enter with no other characters to exit this procedure and calculate the calibration table.
5.) Repeat steps 2 thru 4 until you have finished the calibration for the selected channel.
6.) Use the command *CALRFL,RF-* and repeat the above procedure for the other RF output phase. Make sure you move your oscilloscope probe to the RF- monitoring point.
7.) The calibration data is only in volatile memory when the procedure is completed, enter the command *SAVE* to store the data in non-volatile memory. After entering the *SAVE* command, the RF mega will load the calibration table every time the system starts up.

MIPS App Interface

Interfacing with the MIPS host application and using the control panel feature is the best way to use this system and allows monitoring all the critical parameters. Below is a screen capture of the RFmega's parameters displayed on a control panel:



The control panel configuration file commands use to generate the RFmega dialog are listed below:

```
GroupBox,RFmega,200,260,200,50
Ccontrol,Enable,RFmega,CheckBox,SENA_FALSE,SENA_TRUE,GENA_TRUE_FALSE,,10,20
STEP,10
Ccontrol,Drive,RFmega,LineEdit,GDRV,SDRV,,%,10,45
Ccontrol,Vpp SP,RFmega,LineEdit,GRFV,SRFV,,V,10,70
STEP,1000
```

```
Ccontrol,Freq,RFmega,LineEdit,GFREQ,SFREQ,,Hz,10,95
STEP,10
Ccontrol,RF+,RFmega,LineEdit,,,GRFAVP,Vpp,10,120
Ccontrol,RF-,RFmega,LineEdit,,,GRFAVN,Vpp,10,145
Ccontrol,Power,RFmega,LineEdit,,,GPWR,W,10,170
Ccontrol,Closed loop,RFmega,CheckBox,SMODE_FALSE,SMODE_TRUE,GMODE_TRUE_FALSE,,10,205
Ccontrol,Tune,RFmega,Button,,TUNE_TRUE,,,20,230
GroupBoxEnd
```

Updating the RFmega Firmware:

The RFmega firmware (the software that is programmed into the microcontroller inside the RFmega) can be updated in the field. The procedure in this section outlines two options for updating the firmware. There are two RFmega hardware options we manufacture. An option with both a USB and an Ethernet interface and a second model with only a single RS232 DB9 interface. If your RFmega is the model with only the single RS232 DB9 interface you can only use the first option.

Please note; programming the firmware will rest the calibration to the factory default settings so you will need to re-calibrate the system when complete.

Option 1

The RFmega uses an Adafruit Itsybitsy M4 microcontroller module. This module has a USB connector that is used to program the firmware. This programming option requires this USB port to be connected to your computer. If you RFmega has a USB connector on the front panel then you can just use this connection. If not you will need to use a USB cable and remove the RFmega front panel to make this connection.

The RFmega power does not need to be applied for this procedure. With the USB connection between your computer and the RFmega microcontroller complete and the front panel removed you will next need to press the small reset button on the Itsybitsy controller twice rapidly. This will start the microcontroller in the bootloader mode and a USB drive will appear on your computer called ITSYM4BOOT. You will also notice a red LED on the microcontroller strobing on and off slowly.

The next step is to copy the new firmware file with the .uf2 extension to the USB drive. This will complete the procedure.

Option 2

This option will use the MIPS host application to update the firmware. This procedure can only be performed using RFmega model that has a USB connector. You do not need to remove the RFmega front panel to perform this procedure.

Power on the RFmega and connect to it using the MIPS host application. Make sure the RF outputs are disabled. Under the tools menu you will find a Program RFmega option. Select this option and you will then be asked for the .bin file to program. After to select the .bin file programming procedure will start.

You can find the most current version of the MIPS application on our google drive here:
https://drive.google.com/drive/folders/0B3IchPRNYqYIcjZhdGFVMVR1VzQ?resourcekey=0-qTdIhau9LysBytglIla-yg&usp=sharing

RS232 Host Computer Interface:

All commands are ASCII text.
All commands sent to the RFmega are terminated with a carriage return (CR) or line feed (LF) character.
All messages sent from the RFmega are terminated with a CR and LF.
After a command is received (after the CR or LF is received) RFmega will respond with an ASCII ACK (0x06) then a CR or an ASCII NAK (0x15) then a "?" character and a finally a CR. A NAK indicates the command or its arguments was not understood.
When a NAK is sent the "?" character is sent to inform users communicating with a terminal emulator that the command was not understood.

Commands:

GVER
: Returns the current version string.

GERR
: Returns an integer error code indicating the reason for the last NAK. This value is never cleared and always indicates the communications last error.

GNAME
: Returns a string that contains the name of the RFmega system.

SNAME,Name
: This command will set the RFmega system name. The Name value is a user defined string. Make sure you issue a Save command to save the changes to flash memory or this name will be lost when power is cycled.

SAVE
: This command will save the RFmega controller parameters to the flash memory.

RESTORE
: This command will read the current settings from the flash memory.

RESET
: This command will cause the RFmega control computer to reboot.

MUTE, Value

This command will stop all messages from the RFmega system to the host computer. The received commands will still be processed but no response will be issued. The value parameter turns this mode on and off where:

Value = OFF to turn the responses on and ON to silence the RFmega system.

ECHO,TRUE or FALSE

Setting ECHO true will cause RFmega to include the command with all responses.

GCMDS

This command will cause the RFmega system to display a list of valid commands that it understands.

THREADS

The RFmega system uses a threading control and each module runs on its own Thread as well as a number of threads for control functions. This command displays all the threads and few key parameters. This command is used to monitor the system and check status, mainly a diagnostic tool.

STHRDENA,Thread name,TRUE or FALSE

This command allows you to enable or disable a Thread. This is done by using the thread name and defining it as TRUE or FALSE. If set to FALSE then the thread will stop. This is development and diagnostic tool.

SENA,TRUE or FALSE

This command will enable or disable the RF outputs. TRUE will enable and FALSE will disable.

GENA

This command will return the enable status, TRUE or FALSE.

SFREQ,Freq

This command will set the RF frequency to the value defined my Freq, in Hz.

GFREQ

This command returns the current RF frequency

SDRV,Value

This command defines the drive level for the RFmega. The drive level (0 to 100) defines how much power is sent to the output amplifier in the RFmega. This drive level affects the RF output, the higher the drive level the higher the output voltage.

GDRV

This command returns the current RF drive level.

SRFV,Value

This command defines the desired (or setpoint) peak to peak voltage on each output of the RFmega. This command only applies when the RFmega mode has been set to closed loop.

GRFV

This command returns the current peak to peak desired (or setpoint) voltage.

GRFAVP

This command returns the current peak to peak output voltage on the RF+ output.

GRFAVN

This command returns the current peak to peak output voltage on the RF- output.

GDRVV

This command returns the drive voltage sent to the RF amplifier in the RFmega.

GDRVI

This command returns the current in amps that is being consumed by the RFmega.

GPWR

This command returns the power in watts that is being consumed by the RF amplifier in the RFmega. Note, this power is dissipated in the RF amplifier, the coil and some of this power is dissipated in the load.

SGAIN,Value

This command allows you to set the gain of the closed loop control mode. This is a unitless value that defines the closed loop performance.

GGAIN

This command returns the current closed loop gain setting.

SMODE,TRUE or FALSE

This command sets the RFmega mode. If TRUE the system is in closed loop mode, in this mode the firmware automatically adjusts the drive level to hold the output voltage at the value defined by the SRFV command. If FALSE the drive level is manually set by the SDRV command.

GMODE

This command returns the current mode.

SMDRV,Value

This command sets the maximum drive level that is allowed. This is a command that allows you to set an upper limit to protect your system.

GMDRV

This command returns the current maximum RF drive limit setting.

SMPWR,Pwr

This command allows you to define a maximum RF power level that is permitted. If the power exceeds this level the firmware will automatically reduce the drive.

GMPWR

This command returns the current maximum power setting.

CALRFL,RF+ or RF-

This command is used to calibrate the RF+ and RF- output level monitors. This is a factory setup command and should not be used.

Ethernet commands:

The RFmega system automatically detects the Ethernet module on power up and enables the following commands if the module is found. These commands should be set using the USB host interface.

GEIP

This command will return the current IP address used by the Ethernet module.

SEIP,<xxx.xxx.xxx.xxx>

This command allows you to define the Ethernet module's IP address.

GEPORT

This command returns the port number used for telnet connections to the MIPS system using the Ethernet interface.

SEPORT,<Port>

This command sets the port number used for telnet connections to the MIPS system, the default value is 2015 and this is the value expected by the MIPS host computer application.

Where:

Port = TCP/IP port number

GEGATE

This command will return the gateway IP address used by the Ethernet module.

SEGATE,<xxx.xxx.xxx.xxx>

This command allows you to define the gateway IP address.

GESNIP

This command will return the subnet IP address used by the Ethernet module.

SESNIP,<xxx.xxx.xxx.xxx>

This command allows you to define the subnet IP address.

Warrantee

GAA Custom Electronics, LLC warrantees the MIPS system to be free from defects in materials and workmanship and will repair or replace the unit for a period of one year. This warrantee assumes the system is operated in compliance with the procedures and recommendation outlines in this document. GAA Custom Electronics, LLC will also provide free phone support and firmware bug fixes for up to one year. The addition of new features is not covered in this warrantee.

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